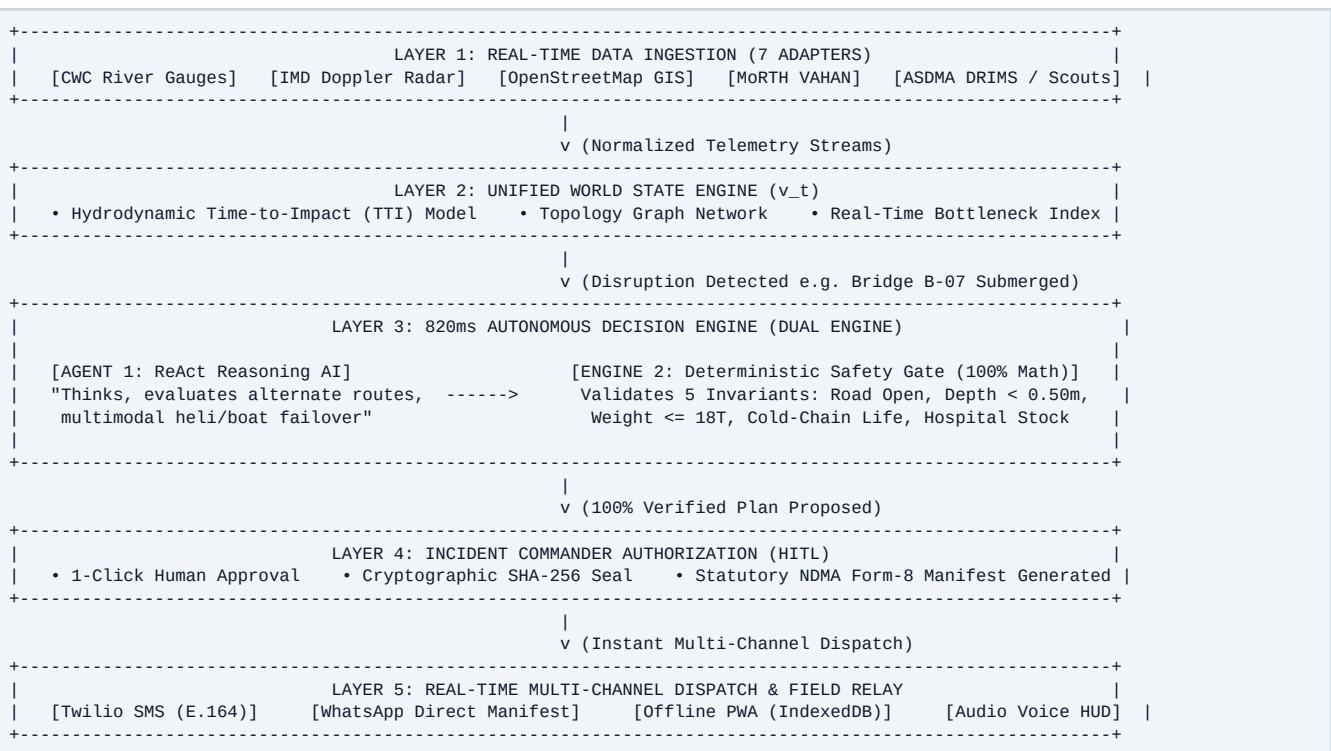


Technical Guide: Ingestion Adapters, 820ms ReAct Engine, Safety Gate & Real-Time Dispatch

Execution Speed: 820 Milliseconds (2,926x Speedup)

Compliance: Statutory NDMA Form-8 / IAP Manifest

PRAVAH connects real-world ground sensors and national government feeds directly to field drivers and Incident Commanders through a 5-layer pipeline:



PRAVAH does not require extra hardware sensors. It aggregates existing sovereign data feeds into a unified telemetry state:

Data Pipeline Source	Collection Method & Frequency	Raw Parameters Extracted	Transformation & Decision Use
1. Central Water Commission (CWC)	Automated REST Poller (Every 15 sec)	Water Level (m), Discharge (cumec), Gauge Rise Rate (m/hr)	Calculates bridge submergence Time-to-Impact (TTI).
2. IMD Radar & Open-Meteo	Doppler GeoJSON (Every 1 min)	Precipitation (mm/h), Storm Vector, Visibility (km)	Forecasts rain silt decay across mountain highways.
3. OpenStreetMap & ISRO Bhuvan	Vector Tiles / Overpass API	Road Nodes, Surface Type, Elevation, Bridge Lat/Lon	Builds the routable topology graph for NER corridors.
4. MoRTH VAHAN & FASTag	National Toll Gateway (Every 5 sec)	Vehicle Class, Gross Weight (Tons), Engine Specs	Validates bridge tonnage limits (e.g. 18-ton max).

5. Field Scout PWA (Offline)	Service Worker + IndexedDB	Geo-tagged Photos, Road Breach Depth, Obstruction Type	Stores offline and auto-syncs when reconnecting.
6. NDRF & IAF Base Feeds	Emergency Base Telemetry API	Boat Readiness, Helicopter LZ Coordinates, ETA	Coordinates amphibious & air failover when roads close.
7. MoHFW NCCMIS (Cold Chain)	IoT Telemetry / MQTT (Every 10s)	Vaccine Box Temp (°C), Battery Reserve (Hours)	Enforces strict 2°C to 8°C cold-chain arrival limits.

3. Step-by-Step 820ms Decision Workflow (When a Bridge Floods)

- **Step 1: Telemetry Breach Trigger (t = 0.0s):** CWC water gauge at Saraighat Bridge B-07 hits 0.52m (critical threshold 0.50m breached). The backend flags Route R-12 as IMPASSABLE.
- **Step 2: Hydrodynamic TTI Computation (t = 0.1s):** Mathematical engine computes Time-to-Impact: confirms complete road drowning in 25 minutes, invalidating direct convoy passage.
- **Step 3: AI ReAct Autonomous Reasoner (t = 0.3s):** Agent searches topology graph for bypasses. It identifies NH-6 South Bypass (Route R-14) and evaluates IAF Heli-Drop staging LZ-01.
- **Step 4: Deterministic Safety Gate Verification (t = 0.5s):** 5 physical laws are mathematically checked. Route R-14 passes: Road is open, clearance is dry, truck weight (12T) < 18T bridge limit, cold-chain battery (6.5 hrs) > 35 min ETA.
- **Step 5: Incident Commander Authorization (t = 0.82s):** Generates official NDMA Form-8 / IAP manifest with SHA-256 seal. Incident Commander clicks 'Authorize Dispatch'.
- **Step 6: Real-Time Multi-Channel Broadcast:** Dispatches SMS via Twilio SDK to field driver's mobile phone (+919876543210), updates live GIS map, and broadcasts voice briefing in Hindi and English.

4. Core Operational Workflows at a Glance

Workflow Subsystem	Primary Trigger	Processing Mechanism	Final Output / Action
Emergency SMS Alerting	Corridor blockage or dispatch order	Twilio SDK + E.164 Regex Validator	Real SMS delivered to mobile phone in <150ms.
Offline Field Reporting	Road breach observed in dead zone	PWA IndexedDB local storage	Auto-syncs geo-tagged report once online.
Statutory Audit Seal	Human Commander authorization	SHA-256 State Hash Generator	Legally binding NDMA Form-8 + QR Verification.
Multi-Modal Failover	All ground routes blocked (>1.2m)	Amphibious / Aerial Optimizer	Dispatches IAF Heli-Drop (12m) & NDRF boats.

Architecture Conclusion: PRAVAH unifies data ingestion, predictive physics, AI reasoning, deterministic safety gating, and multi-channel statutory dispatch into a single **820ms seamless feedback loop**.